

Phase 1 — Azure Infrastructure with Terraform

Harrington Capital plc | Infrastructure Engineer Simulation

Objective

Deploy Harrington Capital's Azure estate using Terraform (IaC). No manual portal clicks. Everything as code. Everything committed to GitHub.

By the end of this phase you will have: - Two Azure Resource Groups (prod and dev) - A Virtual Network with three subnets - Two Network Security Groups with rules - Three VMs (1 Windows Server 2022, 2 Ubuntu 22.04) - All resources tagged and committed to `/terraform` in your repo

BUILD

Step 1 — Authenticate to Azure

```
az login
az account show
```

Note your Subscription ID — you will need it.

Step 2 — Prepare your tfvars

```
cd harrington-infrastructure-operations/terraform
cp terraform.tfvars.example terraform.tfvars
```

Edit `terraform.tfvars` and set a secure admin password (min 12 chars, upper + lower + number + symbol).

Never commit terraform.tfvars — it is in `.gitignore`.

Step 3 — Generate an SSH key pair (for Linux VMs)

```
ssh-keygen -t rsa -b 4096 -f ~/.ssh/id_rsa -N ""  
cat ~/.ssh/id_rsa.pub
```

The public key path `~/.ssh/id_rsa.pub` matches the default in `terraform.tfvars.example`.

Step 4 — Initialise Terraform

```
terraform init
```

Expected output includes:

```
Terraform has been successfully initialized!
```

Step 5 — Complete the TODO in main.tf

Open `/terraform/main.tf` and find the TODO comment in `azurerm_network_security_group.mgmt`.

Add the missing security rule: - Name: `allow-rdp-mgmt-inbound` - Priority: 100 - Direction: Inbound - Access: Allow - Protocol: Tcp - Source port range: `*` - Destination port range: `3389` - Source address prefix: `10.0.3.0/24` (management subnet only) - Destination address prefix: `*`

Step 6 — Plan

```
terraform plan -out=tfplan.bin
```

Review the plan output. You should see **18 resources to add**.

Step 7 — Apply

```
terraform apply tfplan.bin
```

Type `yes` when prompted (if not using the saved plan).

Wait approximately 5-8 minutes for all VMs to provision.

Step 8 — Commit your completed main.tf

```
git add terraform/main.tf  
git commit -m "feat(terraform): add RDP NSG rule - Phase 1 complete"  
git push origin main
```

VERIFY

Run these checks after apply completes:

```
# Confirm all outputs populated
terraform output

# Confirm resource group exists
az group show --name rg-harrington-prod --query "{name:name, location:location, tags:tags}"

# List all resources in the RG
az resource list --resource-group rg-harrington-prod --output table

# Confirm VMs are running
az vm list --resource-group rg-harrington-prod --output table

# Get public IPs
terraform output dc01_public_ip
terraform output app01_public_ip
terraform output mon01_public_ip
```

Update `ansible/inventory.ini` with the public IPs from the outputs.

SUBMIT

Your Phase 1 submission must include:

#	Evidence	How to get it
1	<code>terraform plan</code> output	Copy the summary: <code>Plan: X to add, 0 to change, 0 to destroy.</code>
2	<code>terraform apply</code> output	Copy the final <code>Apply complete! Resources: X added.</code> line
3	<code>terraform output</code>	Full output showing all IPs and IDs
4	Azure Portal screenshot	Resource Group <code>rg-harrington-prod</code> showing all resources
5	GitHub commit URL	URL of your commit adding the RDP NSG rule
6	Architecture diagram	Draw.io network topology (see <code>/architecture/README.md</code>)

Anti-fake check: Your submission must include real Azure Resource IDs. These only exist if `terraform apply` ran successfully. Format: `/subscriptions/[guid]/resourceGroups/rg-harrington-prod/providers/...`

Troubleshooting

Error	Fix
Error: A resource with the ID already exists	Run <code>terraform import</code> or destroy and re-apply
Error: Insufficient quota	Reduce VM sizes to <code>Standard_B1s</code> in <code>terraform.tfvars</code>
Error: SSH public key not found	Run <code>ssh-keygen</code> first and verify path in tfvars
Error: authentication failed	Run <code>az login</code> again — token may have expired
Error: admin_password does not meet complexity	Use a password with upper, lower, number, and symbol

Architecture Diagram Task

Open Draw.io (<https://app.diagrams.net>).

Build `hc-network-topology.drawio` using the reference in `/architecture/README.md`.

Include: - Azure VNet boundary box - Three subnets (app, data, mgmt) with CIDR labels - Three VMs with private IP labels - NSG annotations on app and mgmt subnets - Internet boundary at the top

Export as PNG (File → Export → PNG, 1920x1080).

Commit both files:

```
git add architecture/hc-network-topology.drawio architecture/hc-network-topology.png
git commit -m "docs(architecture): network topology diagram - Phase 1"
git push origin main
```